

asks for an energy

VQE

lowest energy of
a quantum system

a search turns
the dials

molecules,
materials

imaginary time

the same target,
without a search

the physics picks
the next step

when the search
itself stalls

asks for a choice

QAOA

best of exponentially
many choices

a search sets
p pairs of angles

routing,
scheduling

annealing

the same target,
in one slow sweep

no dials at all:
just go slowly

large discrete
problems

in each pair, the left method searches for its own settings and the right one lets the physics fix them